

MAT 310, Handout 2: Discrete Probability Models (2.4, 2.5)

- An _____ is the processing of collecting observations.
- The _____ of an experiment, denoted as $\mathbb{S}$, is the set of all possible outcomes.
- Each outcome in $\mathbb{S}$ is called a _____.
- An _____ is a subset of $\mathbb{S}$ consisting of 0, 1, or more sample points.

Example 1. Consider the experiment of flipping 3 coins and observing whether each is heads or tails. What is $\mathbb{S}$?

Example 2. What is $\mathbb{S}$ when rolling 1 die and observing the number?

Example 3. What is $\mathbb{S}$ when rolling 2 dice and observing both numbers?

Example 4. Flip 3 coins. List all the outcomes in the following events:

- (a) $A = \{\text{at least 2 Heads}\}$
- (b) $B = \{1^{\text{st}} \text{ coin flipped is a Head}\}$
- (c) $A \cap B$
- (d) $A \cap \overline{B}$
- (e) $A \cup B$

Discrete Sample Space: finite or countably infinite # outcomes in $\mathbb{S}$

Natural Question: what is the probability of an event occurring?

For event $A \subseteq \mathbb{S}$, $P(A)$ is the probability that A occurs.

Axioms of Probability

1. $P(A) \geq 0$
2. $P(\mathbb{S}) = 1$
3. If $A_1, A_2, \dots$ are pairwise mutually disjoint, i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$, then:

$$P\left(\bigcup_i A_i\right) =$$

Example 5. Roll 1 die. Given the events: A = roll an odd #, B = roll a 4, and C = roll a prime #. Find:

- | | |
|------------|-------------------------|
| (a) $P(A)$ | (d) $P(A \cup B)$ |
| (b) $P(B)$ | (e) $P(A \cup C)$ |
| (c) $P(C)$ | (f) $P(\bar{A} \cap C)$ |

Example 6. Suppose A and B are disjoint events with $P(A) = .3$ and $P(B) = .5$. Find the probability that:

- (a) either A or B occurs
- (b) A occurs but B does not
- (c) both A and B occur

Calculating Probability of an Event: Sample Point Method

Let event $A \subseteq \mathbb{S} = \{E_1, E_2, \dots, E_n\}$, where E_i is sample point i .

If $A = \{E_i, E_j, \dots, E_k\}$, then $A = E_i \cup E_j \cup \dots \cup E_k$, since all sample points are _____. Therefore,

$$P(A) =$$

Example 7. Roll 2 dice. Find the probability of the event $A = \{\text{sum of 2 dice is 7}\}$.

Probability of Events in Equiprobable Sample Space

If sample points are equiprobable, then $P(A) =$

where $n_A = \#$ sample points in A , and $N = \#$ sample points in $\mathbb{S}$.